

Linear Algebra – MAT 2610

Section 5.1 (Eigenvectors and Eigenvalues)

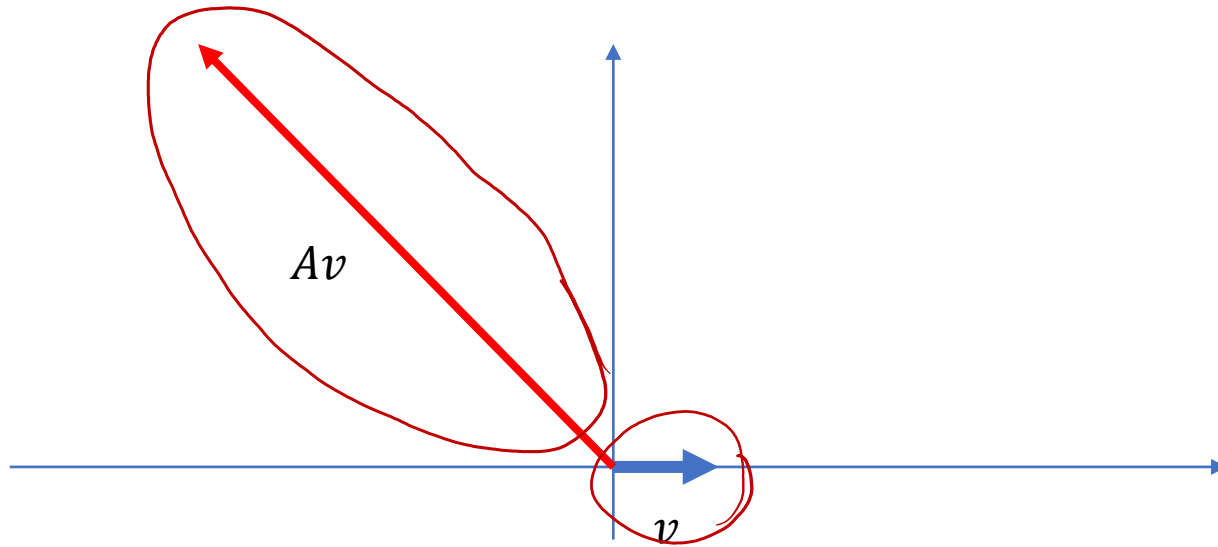
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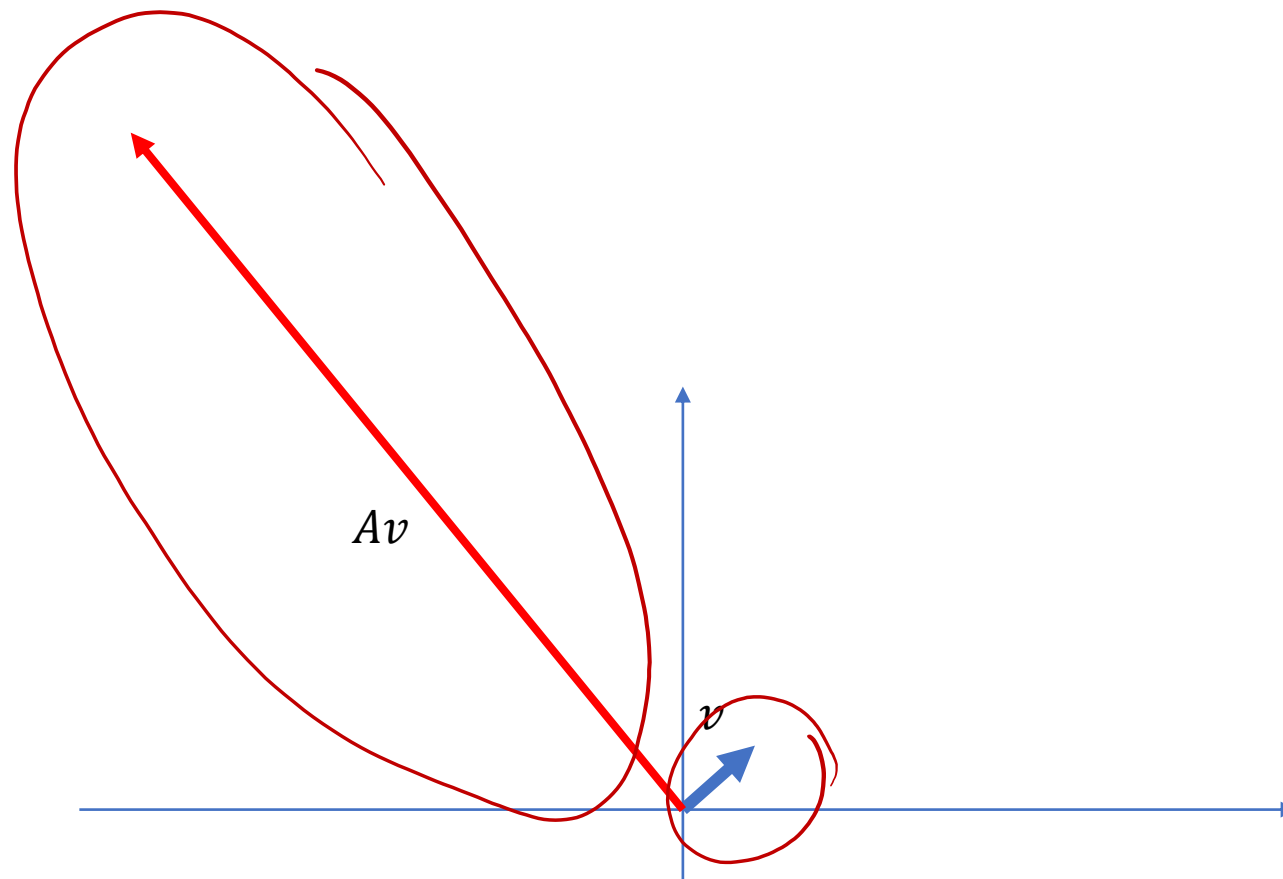
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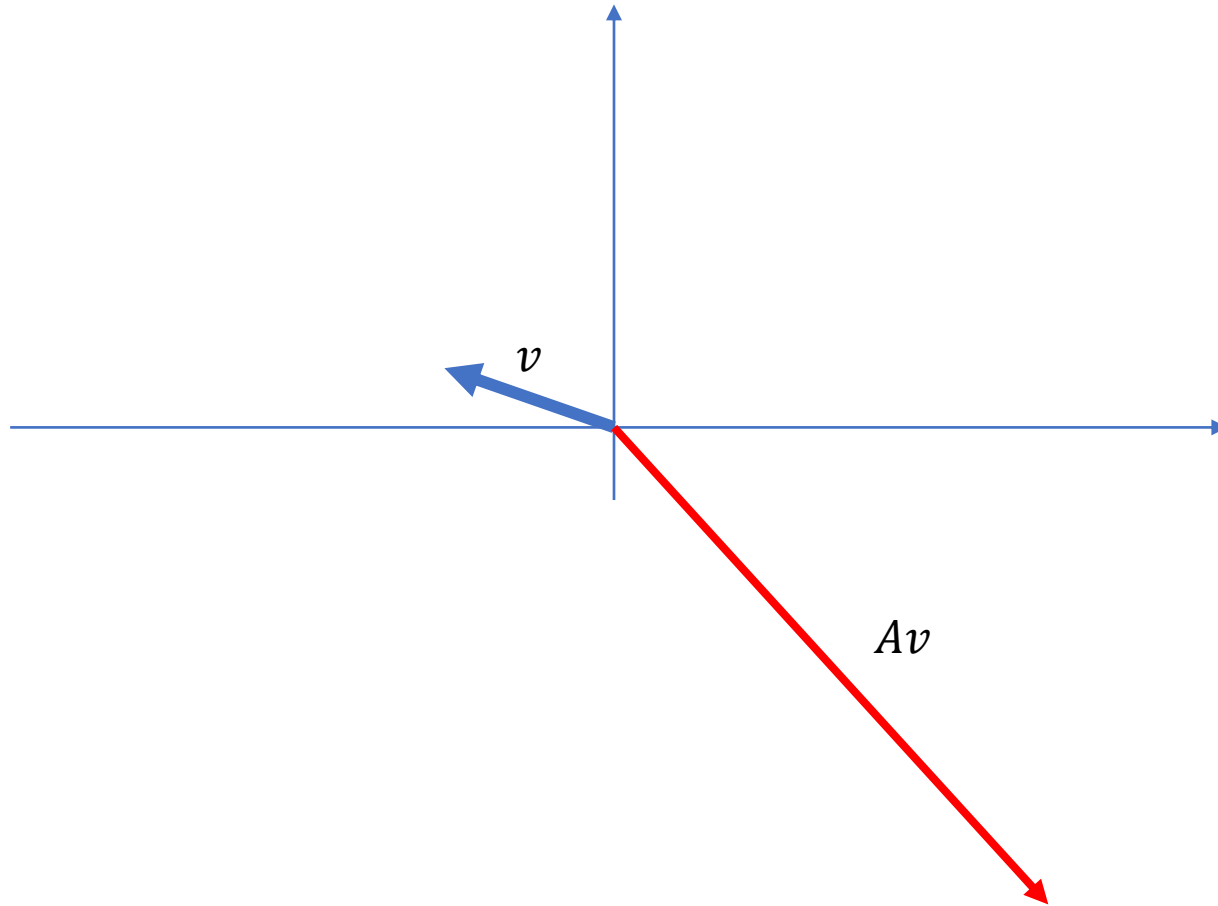
Consider this matrix $A = \begin{bmatrix} -5 & -4 \\ 8 & 7 \end{bmatrix}$, and plot the vectors v and Av for a selection of vectors in R^2 . For example, if $v = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, then $Av = \begin{bmatrix} -5 & -4 \\ 8 & 7 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -5 \\ 8 \end{bmatrix}$



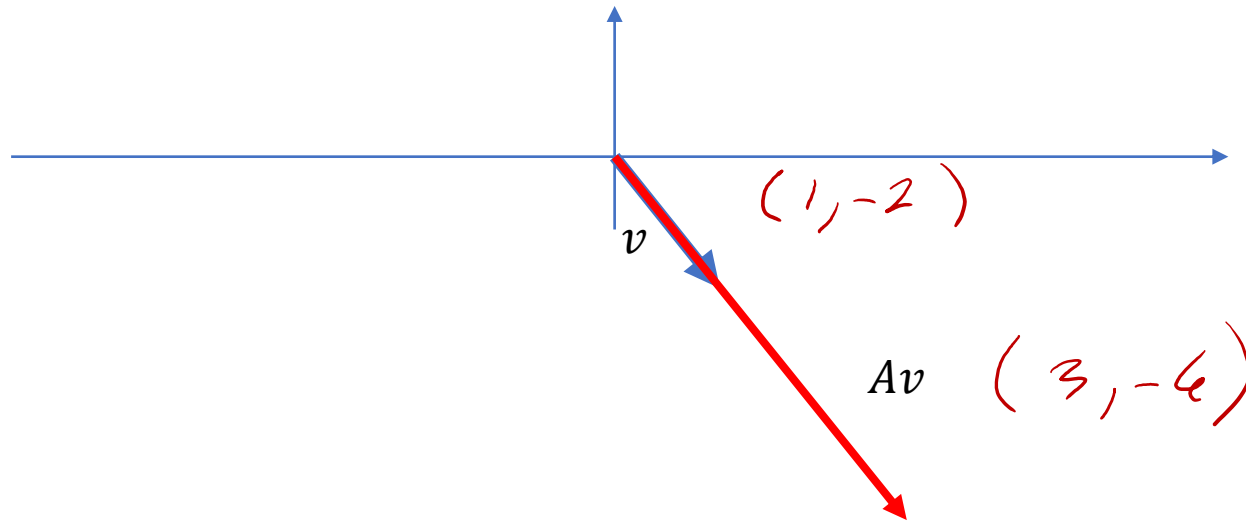
$$\text{If } v = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \text{ then } Av = \begin{bmatrix} -5 & -4 \\ 8 & 7 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -9 \\ 15 \end{bmatrix}$$



$$\text{If } v = \begin{bmatrix} -2 \\ 1 \end{bmatrix}, \text{ then } Av = \begin{bmatrix} -5 & -4 \\ 8 & 7 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ -9 \end{bmatrix}$$



But, if $v = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$, something special happens. When we calculate the value of $Av = \begin{bmatrix} -5 & -4 \\ 8 & 7 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \end{bmatrix} = \begin{bmatrix} 3 \\ -6 \end{bmatrix}$, we see that Av is just a scalar multiple of the v we started with (in particular, $Av = 3v$).



When this happens, we say that the vector $v = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$ is an **eigenvector** of A and the number 3 is an **eigenvalue** of A .



Definition: Let A be an $n \times n$ matrix. A non-zero vector $v \in R^n$ is called an eigenvector of A if $Av = \lambda v$ for some scalar λ . The scalar λ is called the eigenvalue of A that corresponds to v

$$\begin{bmatrix} 4 & 6 & -5 \\ 9 & 7 & -11 \\ 8 & 8 & -11 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} -4 + 12 - 5 \\ -9 + 14 - 11 \\ -8 + 16 - 11 \end{bmatrix} = \begin{bmatrix} 3 \\ -6 \\ -3 \end{bmatrix} = (-1) \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}$$

$$\text{e. vector} = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}$$

$$\text{e. value} = -1$$



Example: The matrix $A = \begin{bmatrix} 1 & 1 \\ 6 & 2 \end{bmatrix}$ has two eigenvalues: $\lambda_1 = -1$ and $\lambda_2 = 4$. Find their corresponding eigenvectors.

$$\begin{bmatrix} 1 & 1 \\ 6 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -x_1 \\ -x_2 \end{bmatrix}$$

$$x_1 + x_2 = -x_1$$

$$6x_1 + 2x_2 = -x_2$$

$$2x_1 + x_2 = 0$$

$$6x_1 + 3x_2 = 0$$

$$2x_1 + x_2 = 0$$

$$x_2 = -2x_1$$

$$\text{Let } x_1 = 1$$

$$\text{then } x_2 = -2$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 6 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 4x_1 \\ 4x_2 \end{bmatrix}$$

$$x_1 + x_2 = 4x_1$$

$$6x_1 + 2x_2 = 4x_2$$

$$-3x_1 + x_2 = 0$$

$$6x_1 - 2x_2 = 0$$

$$x_2 = 3x_1$$

$$\text{Let } x_1 = 1$$

$$\text{then } x_2 = 3$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$





Finding Eigenvalues and Eigenvectors

When finding the eigenvalues and eigenvectors, you solve the equation $Ax = \lambda x$ for vector x and scalar λ . The first step will be to find the values of λ . There are usually multiple options for the values of λ .



$$Ax = \lambda x$$

$$Ax = \lambda I_n x$$

$$x = I_n x$$



$$Ax - \lambda I_n x = 0$$

$$(A - \lambda I_n)x = 0$$

matrix vector



Finding Eigenvalues and Eigenvectors

So, to find eigenvalues and eigenvectors, we need to solve $(A - \lambda I_n)x = 0$. Remember that $x = 0$ is not an eigenvector, even though it satisfies this equation.

That means that $(A - \lambda I_n)x = 0$ has an infinite number of solutions for x .

Recall that $(A - \lambda I_n)x = 0$ has an infinite number of solutions exactly when $A - \lambda I_n$ is not invertible.

That means that the eigenvalues of A are all values λ such that $\det(A - I\lambda) = 0$

Use this fact to find all the eigenvalues $\{\lambda_1, \lambda_2, \dots, \lambda_k\}$, then for each find the eigenvectors



Theorem 1

The eigenvalues of a triangular matrix are the entries on its main diagonal.

$$A = \begin{bmatrix} 3 & 2 & 1 \\ 0 & 1 & 4 \\ 0 & 0 & 2 \end{bmatrix}$$

$$\begin{aligned} \lambda_1 &= 3 \\ \lambda_2 &= 1 \\ \lambda_3 &= 2 \end{aligned}$$

$$\begin{aligned} A - \lambda I &= \begin{bmatrix} 3-\lambda & 2 & 1 \\ 0 & 1-\lambda & 4 \\ 0 & 0 & 2-\lambda \end{bmatrix} \\ \det(A - \lambda I) &= 0 \\ &= \underbrace{(3-\lambda)} \underbrace{(1-\lambda)} \underbrace{(2-\lambda)} = 0 \end{aligned}$$



Example: Find the eigenvalues and eigenvectors of $A = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 2 & -1 \\ 0 & 0 & -1 \end{bmatrix}$

$$\lambda_1 = 1 \quad \lambda_2 = 2 \quad \lambda_3 = -1$$

$$\begin{bmatrix} 1 & -1 & 0 \\ 0 & 2 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\begin{aligned} x_1 - x_2 &= x_1 \\ 2x_2 - x_3 &= x_2 \\ x_3 &= x_3 \end{aligned} \Rightarrow \begin{aligned} -x_2 &= 0 \\ x_2 - x_3 &= 0 \\ 0 &= 0 \end{aligned} \Rightarrow \begin{aligned} x_2 &= 0 \\ x_3 &= 0 \\ x_1 &\text{ free} \end{aligned}$$

$$\text{e. vector} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$



$$\begin{bmatrix} 1 & -1 & 0 \\ 0 & 2 & -1 \\ 0 & 0 & -1 \end{bmatrix} \quad \lambda_2 = 2$$

$$\begin{aligned} x_1 - x_2 &= 2x_1 \\ 2x_2 - x_3 &= 2x_2 \\ -x_3 &= 2x_3 \end{aligned} \Rightarrow$$

$$-x_1 - x_2 = 0$$

$$-x_3 = 0 \Rightarrow$$

$$-3x_3 = 0$$

$$x_3 = 0$$

$$\begin{aligned} \text{let } x_1 &= 1 \\ \text{then } x_2 &= -1 \end{aligned}$$

$$\text{e. vector} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

$$\lambda_3 = -1$$



Theorem 2

If $v_1, \dots, v_r$ are eigenvectors that correspond to distinct eigenvalues $\lambda_1, \dots, \lambda_r$ of an $n \times n$ matrix A , then the set $\{v_1, \dots, v_r\}$ is linearly independent.



Theorem 2 - Proof

If $\{v_1, \dots, v_r\}$ are linearly dependent, then select $p \leq r$ to be the smallest number such that $\{v_1, \dots, v_{p+1}\}$ is linearly dependent

$$c_1 v_1 + \dots + c_p v_p = v_{p+1} (*) \quad \leftarrow$$

We can multiply this by λ_{p+1} to get

$$\underset{\uparrow}{c_1} \underset{\uparrow}{\lambda_{p+1}} v_1 + \dots + \underset{\uparrow}{c_p} \underset{\uparrow}{\lambda_{p+1}} v_p = \underset{\uparrow}{\lambda_{p+1}} v_{p+1} (**) \quad \leftarrow$$



Theorem 2 - Proof

Multiply (*) by A and replace using eigenvalues

$$\begin{aligned} \rightarrow c_1 Av_1 + \dots + c_p Av_p &= Av_{p+1} \\ c_1 \lambda_1 v_1 + \dots + c_p \lambda_p v_p &= \lambda_{p+1} v_{p+1} \end{aligned} \quad \leftarrow$$

Now subtract (**) from this to get

$$c_1(\lambda_1 - \lambda_{p+1})v_1 + \dots + c_p(\lambda_p - \lambda_{p+1})v_p = (\lambda_{p+1} - \lambda_{p+1})v_{p+1} = \underline{0} \quad \leftarrow$$

This shows that $\{v_1, \dots, v_p\}$ is linearly dependent, which is a contradiction.



Definition: Given a particular eigenvalue λ_0 of a matrix A , the **eigenspace corresponding to λ_0** is the set of all solutions to $(A - \lambda_0 I_n)x = 0$ (including the zero vector).

Matrix Vector

Note that this is a null space.



Example: Find the eigenvalues and eigenvectors of $A = \begin{bmatrix} 6 & 1 & 3 \\ 2 & 5 & 3 \\ 2 & 1 & 7 \end{bmatrix}$ and eigenvalue

$\lambda = 4$, find a basis for the corresponding eigenspace

$$A - \lambda I = A - 4I = \begin{bmatrix} 6-4 & 1 & 3 \\ 2 & 5-4 & 3 \\ 2 & 1 & 7-4 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 3 \\ 2 & 1 & 3 \\ 2 & 1 & 3 \end{bmatrix} \leftarrow \text{Find null space}$$

$$\begin{bmatrix} 2 & 1 & 3 \\ 2 & 1 & 3 \\ 2 & 1 & 3 \end{bmatrix} \xrightarrow[r_1 \rightarrow r_3 \rightarrow r_3]{r_1 - r_2 \rightarrow r_2} \begin{bmatrix} 2 & 1 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{array}{l} 2x_1 + x_2 + 3x_3 = 0 \\ x_2, x_3 \text{ free} \end{array}$$

$$\begin{array}{l} x_1 = -\frac{1}{2}x_2 - \frac{3}{2}x_3 \\ x_2 = x_2 \\ x_3 = x_3 \end{array} \Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_2 \begin{bmatrix} -1/2 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -3/2 \\ 0 \\ 1 \end{bmatrix}$$



eigenspace basis $\left\{ \begin{bmatrix} -1/2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -3/2 \\ 0 \\ 1 \end{bmatrix} \right\}$



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